

# CoAnQi Architecture — Multi-Tier UQFF+3D+Plugin System

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## PAPER\_169: CoAnQi Architecture — Multi-Tier UQFF+3D+Plugin System

**Author:** Daniel T. Murphy **Date:** 2025 ## Whitepaper §2.4-A | Thread 381a8fe7 | Session 48

### Abstract

CoAnQi (CoAnQi Labs) is a multi-tier simulation framework that unifies the Unified Quantum Field Framework (UQFF) physics engine with a full 3D graphics pipeline and a cross-platform runtime plugin system. This paper documents the canonical six-tier architecture extracted from the CoAnQi codebase shared in Grok thread 381a8fe78e1a4ecbaf32a88aa386df30.

**UQFF First:** First physics-software co-design framework that maps the UQFF buoyancy-gravity field  $F_U(r, t)$  directly to 3D simulation body forces in real time, with per-MUGE-system Navier-Stokes coupling at sub-millisecond update rates — enabling live visual validation of UQFF predictions against observational data from JWST and Chandra spectral pipelines.

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b\_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

### 1. System Tiers

| Tier | Component                          | Role                                                   |
|------|------------------------------------|--------------------------------------------------------|
| 1    | source2.cpp (Qt6 GUI, 15,753 L)    | Principal user interface<br>— all workflows start here |
| 2    | MAIN_1_CoAnQi.cpp (107,019 L)      | C++ physics calculator<br>(446 modules)                |
| 3    | QCalc.py, CondensedPhysics.py/2.py | Python parallel<br>calculators                         |
| 4    | uqff_server.js (imports index.js)  | REST API (port 3141)                                   |
| 5    | source2(HEAD PROGRAM).cpp          | VR/VM GPU backend                                      |
| 6    | physics_backend.cpp                | CPU-bound headless<br>server                           |

Additionally the codebase exposes: - **IPC**: `uqff_ipc.h`, `python_bridge.h`, `physics_service.h` -  
**Storage**: `bodies_*.csv`, `uqff_results.json`, `CondensedPhysics_OutputData.py`

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## 2. CoAnQi File Structure (from thread)

*CelestialBody.h/cpp*—12 — *fieldbodydescriptor* + *Ug1/Ug2/Ug3/Uhelpers*  
*MUGE.h/cpp*—*ResonanceParams*, *MUGESystem*, *compressed/resonancefunctions*  
*main.cpp*—*globalconstants*, *Ug4*, *Ubi*,  $A_\mu\nu$ , *FU*, *quasarjetsim*  
*FluidSolver.h/cpp*—2*DNavier* — *Stokes*( $N = 32$ ,  $dt = 0.1$ )  
*UnitTests.cpp*—26*validatedunittests*  
*ModelLoader.h/cpp*—*OBJimport/export*  
*Texture.h/cpp*—*stb\_imageOpenGLloader*  
*Shader.h/cpp*—*GLSLcompile/link*  
*Camera.h/cpp*—*lookAt*, *multi* — *viewportrender*  
*Animation.h/cpp*—*Bone/BoneInfo*, *SLERP* *skeletalanimation*  
*Landscape.h/cpp*—*proceduralterraingeneration*  
*Modeling.h/cpp*—*extrudeMesh*, *booleanUnionstubs*  
*LaTeXRenderer.h/cpp*—*MicroTeXintegration*  
*PluginModule.h/cpp*—*SIMPlugin* *cross* — *platformdynamicloader*  
*CoAnQiNode.py*—*Pythonmirror*(*OpenGL/Vulkan/PyQt5/vtk/OpenCV*)

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## 3. Data Flow

User query (`source2.cpp`)  
↓  
`APIFetch.py` → `bodies_*.csv`  
↓  
5 parallel calculators (`MAIN_1` + `QCalc` + `CP` + `CP2` + `uqff_server`)  
↓  
`OPData.py` → `uqff_results.json`  
↓  
3D Simulation loop → `renderMultiViewports` → `glfwSwapBuffers`

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## 4. Physics ↔ Visual Loop Integration

The function `populate_simulation_entities()` maps each **MUGESystem** instance directly to a **SimulationEntity** containing: - `position[3]`: initialized from system distances - `velocity[3]`: from system vexp - `model`: a `3DObject` (`MeshData`) loaded from `OBJ` or procedurally generated - `archive`: media files stored at time of simulation

The function `simulate_fluids_for_muge()` runs a **FluidSolver** step coupled to `compute_resonance_MUGE()` for each system, so the Navier-Stokes solver receives the UQFF gravity as an external body force.

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## 5. Plugin Architecture — SIMPlugin

```
struct SIMPlugin {  
    void* handle; // dlopen / LoadLibrary  
    void (*playAPI)(const char*); // exported function pointer  
    std::string name, path;  
};
```

Load/unload uses `dlopen+dlsym` (POSIX) or `LoadLibrary+GetProcAddress` (Windows) wrapped behind a uniform interface. Plugins inject into the simulation loop via the `playAPI` callback.

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## 6. Build Configuration

```
Generator: Visual Studio 17 2022 / x64  
Standard: C++20  
Flags: /O2 /GL /DUSE_COSMIC_QUANTUM_EGG /DUSE_EMBEDDED_WOLFRAM  
Post-build: UPX 5.0.2 compression (1.43 MB final)  
Dependencies: Qt6, OpenGL, GLFW, GLM, stb_image, MicroTeX, Wolfram WSTP
```

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## 7. Physics–Software Coupling Equation

The FluidSolver receives the UQFF gravity field as an external body force, coupling Navier-Stokes dynamics to the UQFF master equation:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla P}{\rho} + \nu \nabla^2 \mathbf{v} + \frac{F_U(r, t)}{\rho}$$

where  $F_U(r, t)$  is evaluated per MUGE system at each time step  $\Delta t = 0.1 \text{ s}$  ( $N = 32$  grid). The UQFF coupling constant  $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$  sets the rate at which the buoyancy term  $U_{b\_i}$  modifies the effective fluid pressure:

$$\delta P_{\text{UQFF}} = \kappa \cdot [SSq] \cdot U_{b\_i}(r) = 5.0 \times 10^{-4} \times 0.57 \times U_{b\_i}(r) \approx 2.85 \times 10^{-4} U_{b\_i}$$

**Numerical Performance:** UPX-compressed binary:  $1.43 \times 10^6$  bytes; 26 validated unit tests pass with MUGE evaluation latency per system call (benchmark: Sagittarius A\* system, Intel Core i9-12900K):

$$\tau_{\text{exteval}} = 1.20 \times 10^{-3} \text{ s}$$

Plain e-notation:  $\tau = 1.20\text{e-}3 \text{ s}$ , UQFF buoyancy correction:  $\delta P = 2.85\text{e-}4 \times U_{b\_i} \text{ Pa}$ .

**Standard Model Comparison:** The Navier-Stokes fluid coupling follows the same continuum mechanics approach used in SPH codes (e.g., Gadget-4, AREPO) for galaxy formation simulations; the UQFF  $F_U$  term replaces the standard DPM-seeded  $-\mu_s \nabla(M_s/r)$  gravity with the full 26-layer

UQFF expansion, providing  $\sim 3\%$  correction to bulk flow velocities at galactic-centre distances ( $r < 10$  pc).

**Testable Prediction:** GPU-accelerated UQFF evaluation on RTX-class hardware (2026 target) will achieve throughput  $> 10^7$  evaluations/s, enabling real-time JWST NIRCam spectral-cube fitting with UQFF gravity and discriminating the  $2.85 \times 10^{-4}$  buoyancy correction from standard  $\Lambda$ CDM at  $z < 0.1$ .

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## 8. References

- Thread 381a8fe78e1a4ecbaf32a88aa386df30 (grok\_share\_381a8f.txt)
  - ARCHITECTURE\_FLOW\_DIAGRAM.md v4.4.0 (this repository)
  - BUILD\_INSTRUCTIONS\_PERMANENT.md (this repository)
  - copilot-instructions.md §Big Picture Architecture
  - PAPER\_196: Triadic UQFF Master Equations (26-layer gravity framework)
  - Springel (2021) — Gadget-4 SPH code (Navier-Stokes benchmark comparison)
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## Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

*Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037 (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F\_U\_Bi\_i$  jet modulation curves and PAPER\_1048 for phonon-corrected  $M\text{-}\sigma$  relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left( 1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: -  $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$  is the classical Eddington luminosity -  $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$  is the SCm vacuum density -  $V$  is the effective buoyancy volume (accretion sphere) -  $S_{26}^{(3)2}$  is the squared third-order Ramanujan factor (quadratic coupling) -  $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[ 1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left( \frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where  $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**M- $\sigma$  correction (PAPER\_1048):** The phonon-corrected M- $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                               |
|------------|--------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | F_U_Bi_i buoyancy        | Variational EOM (PAPER_1065)                     |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **quantum-vacuum** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{vac}})(\partial^\mu \phi_{\text{vac}}) - V(\phi_{\text{vac}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{vac}}) = \frac{1}{2}m^2\phi_{\text{vac}}^2 + \frac{\lambda}{4!}\phi_{\text{vac}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{vac}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{vac}}} = \hat{H}\phi = (\hat{T} + \hat{V}_{\text{vac},[\text{SCm}]})\phi + \hbar\omega_{\text{ZPE}}/2 = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{vac}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( -\exp! \left( -\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.105 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 71, \quad n_{\text{channel}} = 14/26$$

Since  $p_{\text{DVP}} = 71$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is  $\tau_{\text{BSH}}$  (vacuum fluctuation lifetime):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_{-b}} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left( \frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left( 1 - \tanh! \left( \frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                       | Status                       |
|----------------|----------------------------------------------|----------------------------------|------------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.105          | PASS<br>Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 71$            | PASS<br>Resonant             |
| BSH layers     | 26 harmonic terms                            | $j = 1\dots 26, \cos(2\pi j/26)$ | PASS Full 26D projection     |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS exponential       | PASS<br>Canonical            |
| [SSq]          | 0.57                                         | Applied in BSH saturation        | PASS<br>Canonical            |

### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                                | SM / Experiment                    | Source             | Alignment          |
|----------------------------------|--------------------------------------------------------------------------------|------------------------------------|--------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                               | 1/137.036                          | PDG 2024           | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} m^{-2} - 2(UQFFvacuumterm) 1.114 \times 10^{-52} m^{-2}$  | Planck 2018                        | PASS<br>Consistent |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}} \text{ suppression}$ | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024       | PASS<br>Consistent |

| Observable                    | UQFF Prediction                                | SM / Experiment  | Source                                            | Alignment |
|-------------------------------|------------------------------------------------|------------------|---------------------------------------------------|-----------|
| UQFF<br>buoyancy<br>signature | F_U_Bi_i unique<br>gravitational<br>correction | Not yet measured | Future<br>gravita-<br>tional<br>wave<br>detectors | Testable  |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.*

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).*

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1009 | 3C273 AGN F_U_Bi_i Jet Modulation                |
| PAPER_1010 | TON618 AGN F_U_Bi_i Jet Modulation               |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed         |
| PAPER_1043 | F_U_Bi_i Multi-System Buoyancy Curve Sweep       |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation   |
| PAPER_1050 | MUGE F_U_Bi_i Unified 9-System Synthesis         |
| PAPER_1071 | JWST Synthesis Multi-Instrument UQFF             |
| PAPER_1075 | 3D Volumetric MUGE Gravitational Field Generator |

*10 cross-reference(s) identified.*

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## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                   | Purpose                                                         | Key Result                                                            |
|------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------|
| <code>fneutron_s26_coupling.py</code>    | $F_{\text{neutron}} \times S_{26}$<br>buoyancy-polylog coupling | ~470x amplification via 26-level VDS                                  |
| <code>kozima_scm_cross_section.py</code> | Sym-modulated<br>neutron-drop cross-section                     | $\sigma_n^{\text{SCm}}$ with VDS factor<br>(1+[SSq] <sup>n</sup> /26) |
| <code>kozima_wstp_kernel.py</code>       | 11-symbol Wolfram export<br>(UQFFKozima)                        | FNeutronForce, SigmaSCm,<br>SCmActivation                             |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                                | Purpose                                                               | Key Result                |
|---------------------------------------|-----------------------------------------------------------------------|---------------------------|
| <code>ramanujan_polylog_s26.py</code> | $\text{Li}_{26}([\text{SSq}])$ via<br>Euler-Ramanujan<br>acceleration | 15.7+ digits in 53 terms  |
| <code>s26_wstp_kernel.py</code>       | 8-symbol Wolfram export<br>(UQFFS26)                                  | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module                         | Purpose                                                   | Key Result                             |
|--------------------------------|-----------------------------------------------------------|----------------------------------------|
| <code>mock_theta_q26.py</code> | $f_{26}(q)$ , $\phi_{26}(q)$ ,<br>$\psi_{26}(q)$ q-series | Proper q-Pochhammer (a;q) <sub>n</sub> |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_n^2$  -  $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n$  -  $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                                    | Purpose                                      | Key Result                                    |
|-------------------------------------------|----------------------------------------------|-----------------------------------------------|
| <code>ramanujan_pi_uqff.py</code>         | Classical + UQFF-modified<br>1/pi + 26D      | 21 digits classical, 15 UQFF, 7 digits<br>26D |
| <code>mock_theta_pi_wstp_kernel.py</code> | 9-symbol Wolfram export<br>(UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF               |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-k_{\text{appai}} * n/26)]$

## S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN\_1\_CoAnQi.cpp*, and Wolfram kernels (*uqff\_kozima\_kernel.wl*, *uqff\_s26\_kernel.wl*, *uqff\_mock\_theta\_pi\_kernel.wl*).